

P A T E N T A P P L I C A T I O N S P E C I F I C A T I O N

Constraint-Aware AIoT Power Grid Self-Healing and Autonomous Fault Recovery System Using Digital Twin, Multi-Modal Anomaly Detection, Topology-Aware Fault Localization, Cascading Risk Prediction, and Recovery Optimization

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ABSTRACT OF THE DISCLOSURE

A system and method for autonomous self-healing of electrical power distribution networks comprising: a virtual power grid simulation engine **110** configured to model electrical topology and compute real-time AC/DC power flow; a virtual Internet of Things (IoT) sensor network **120** generating multi-modal electrical measurements including voltage, current, frequency, active power, reactive power, and temperature; a multi-layer anomaly detection module **140** combining rule-based electrical constraint monitoring with isolation forest machine learning; a topology-aware fault localization engine **160** utilizing graph-theoretic analysis of the grid network to identify faulted components; a cascading failure risk predictor **170** that estimates secondary failure probabilities through graph propagation modeling; a constraint-aware recovery optimization engine **180** that generates candidate recovery configurations, validates each against hard electrical constraints including thermal limits, voltage bounds, transformer capacity, network connectivity, and critical load preservation, and selects an optimal feasible recovery plan; and a closed-loop verification module **195** that re-simulates power flow post-recovery and iterates if stability is not achieved. The system operates as a software-based digital twin **199** enabling reproducible experimentation of grid self-healing without physical hardware.

TECHNICAL FIELD OF THE INVENTION

The present invention relates generally to electrical power grid monitoring, fault detection, and autonomous recovery systems. More particularly, the invention relates to an integrated Artificial Intelligence of Things (AIoT) platform that combines virtual sensor infrastructure, multi-modal anomaly detection, topology-aware fault localization, cascading failure prediction, and constraint-aware recovery optimization to enable autonomous self-healing of electrical power distribution networks.

BACKGROUND OF THE INVENTION

Electrical power grids are among the most critical and complex engineered systems in existence, comprising interconnected generators, transformers, transmission lines, substations, and millions of

end-user loads. Maintaining grid stability requires continuous monitoring of electrical parameters and rapid response to faults.

Conventional Supervisory Control and Data Acquisition (SCADA) systems and Energy Management Systems (EMS) provide real-time monitoring of grid parameters. These systems typically employ fixed-threshold alarms that trigger when measurements exceed predefined limits. While effective at detecting gross abnormalities, such systems suffer from significant limitations:

First, traditional monitoring systems detect anomalies but lack the capability to classify fault types, localize fault positions within the grid topology, or predict cascading consequences. An operator receives an alarm indicating abnormal voltage at a particular bus but must manually determine whether the cause is a line failure, transformer overload, generator trip, or sensor malfunction.

Second, existing systems do not model cascading failure propagation. A single component failure can redistribute power flows across the network, overloading adjacent components and triggering sequential trips. The 2003 Northeast Blackout and the 2012 India Blackout demonstrate that cascading failures can escalate from a single local fault to a continental-scale outage within minutes.

Third, recovery actions in conventional systems are either manually determined by human operators or based on pre-programmed lookup tables. Neither approach accounts for the real-time electrical state of the grid. An automated switching action that appears correct in isolation may violate thermal limits on alternate paths, creating new faults while resolving the original one.

Fourth, the proliferation of IoT sensors introduces data reliability challenges. Sensor failures, communication dropouts, and corrupted measurements can produce readings indistinguishable from genuine grid faults. Acting on erroneous sensor data leads to unnecessary and potentially dangerous switching operations.

Fifth, experimentation with grid self-healing algorithms on physical infrastructure is impractical due to safety risks, regulatory constraints, and the inability to reproduce fault conditions. There exists no comprehensive software environment that faithfully simulates the complete detect-diagnose-localize-predict-recover-verify cycle.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a formal block diagram of the overall constraint-aware AIoT power grid self-healing architecture showing data flow between subsystems.

FIG. 2 is a detailed flowchart of the closed-loop autonomous self-healing process execution sequence from sensor ingest to verified recovery.

FIG. 3 is an electrical single-line schematic diagram of the virtual power grid topology including power plant, substations, transmission lines, transformers, and critical loads.

FIG. 4 is a architectural schematic of the multi-layer anomaly detection, fault classification, and topology-aware localization subsystem.

FIG. 5 is a process flow diagram of the constraint-aware recovery optimization engine illustrating candidate generation, hard constraint filtering, and multi-objective scoring.

FIG. 6 is a schematic layout diagram of the digital twin human-machine interface and control center dashboard.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

1. Integrated System Architecture (FIG. 1)

Referring to **FIG. 1**, the system **100** comprises a virtual power grid simulation module **110**, a virtual IoT sensor network **120**, a data ingestion and validation layer **130**, a multi-layer anomaly detection engine **140**, a fault classification module **150**, a topology-aware fault localization engine **160**, a cascading failure risk predictor **170**, a constraint-aware recovery optimization engine **180**, an action executor **190**, a post-recovery verification module **195**, and a digital twin visualization interface **199**.

The virtual power grid simulation module **110** computes real-time AC/DC power flow solutions using Newton-Raphson iterations, determining bus voltage magnitudes, phase angles, line active/reactive powers, and equipment utilization percentages.

The virtual IoT sensor network **120** generates continuous multi-modal telemetry comprising voltage (\$V\$), current (\$I\$), frequency (\$f\$), active power (\$P\$), reactive power (\$Q\$), temperature (\$T\$), power factor (\$PF\$), breaker status, and line status.

2. Closed-Loop Self-Healing Process (FIG. 2)

Referring to **FIG. 2**, the self-healing process **200** executes autonomously. Step **201** ingests live telemetry. Step **202** runs rule-based and ML anomaly checks. Step **203** cross-validates sensor channels to reject sensor hardware glitches. Step **204** classifies the fault into one of twelve fault categories. Step **205** computes graph-theoretic component localization. Step **206** calculates severity. Step **207** estimates cascading failure probability on adjacent branches.

Steps **208–213** isolate the faulted branch, synthesize multiple recovery candidates, enforce physical constraint gates, score feasible candidates, and select the optimal plan. Step **214** commands breaker actions. Step **215** re-computes power flow. Steps **216–217** confirm stability or iterate.

CLAIMS

What is claimed is:

1. A system for autonomous self-healing of an electrical power grid, comprising:

- (a) a power grid simulation engine **110** configured to model an electrical network topology comprising buses, generators, transmission lines, transformers, loads, and switching devices, and to compute power flow solutions producing voltage, current, and loading values for each network component;
- (b) a virtual sensor network **120** comprising a plurality of software-simulated sensor agents, each associated with a grid component and configured to generate periodic multi-modal measurements including voltage, current, frequency, active power, reactive power, and temperature;
- (c) a multi-layer anomaly detection module **140** comprising a rule-based electrical constraint checker, a machine learning anomaly detector, and a temporal pattern analyzer, configured to identify abnormal grid conditions from the sensor measurements;
- (d) a fault classification module **150** configured to categorize detected anomalies into a plurality of fault types including transmission line failure, transformer overload, voltage instability, frequency disturbance, generator failure, and sensor malfunction;
- (e) a topology-aware fault localization engine **160** configured to represent the grid as a graph and compute fault probability rankings for grid components based on sensor correlations and network topology;
- (f) a cascading failure risk predictor **170** configured to simulate power flow redistribution following a fault and estimate secondary failure probabilities;
- (g) a constraint-aware recovery optimization engine **180** configured to generate candidate recovery configurations, validate each against hard electrical constraints, reject infeasible candidates, and select an optimal feasible recovery plan; and
- (h) a closed-loop verification module **195** configured to re-simulate power flow after recovery execution and iterate recovery planning if stability criteria are not satisfied.

2. The system of claim 1, wherein the hard electrical constraints comprise line thermal capacity limits, transformer loading limits, bus voltage bounds, network connectivity requirements,

generator capacity limits, power balance equations, and critical load preservation requirements.

3. The system of claim 1, wherein the recovery optimization engine computes an objective function $J(P) = w_1 \cdot \text{RestoredLoad} - w_2 \cdot \text{CascadeRisk} - w_3 \cdot \text{SwitchingCost} - w_4 \cdot \text{OverloadPenalty}$.

4. The system of claim 1, wherein the virtual sensor network generates measurements incorporating baseline operating values, stochastic noise, diurnal load patterns, inter-sensor correlations, and fault-induced deviations proportional to fault severity.

5. The system of claim 1, wherein the anomaly detection module distinguishes genuine grid faults from sensor faults by cross-referencing adjacent graph sensors.

FORMAL DRAWINGS OF THE INVENTION

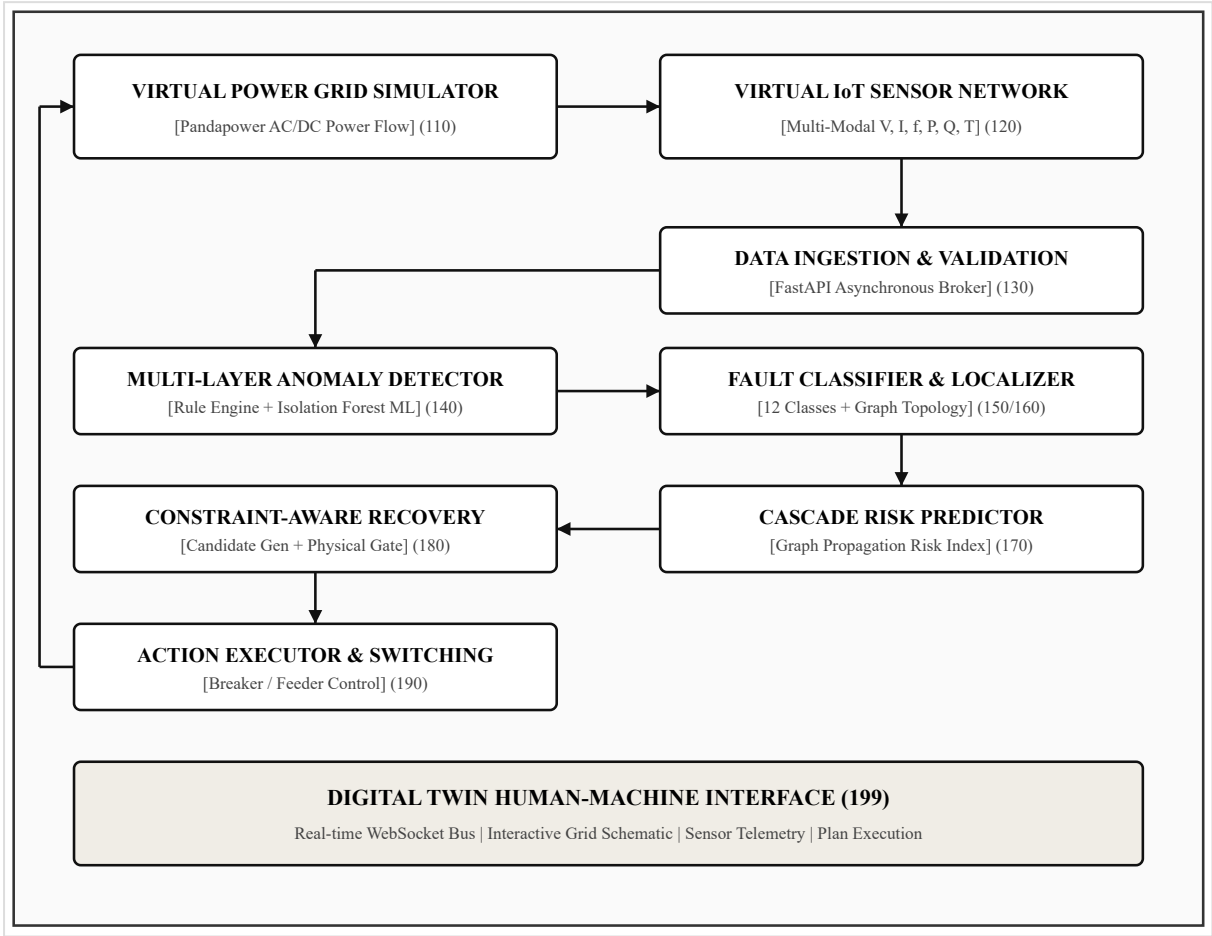


FIG. 1

Overall Constraint-Aware AIoT Power Grid Self-Healing Architecture

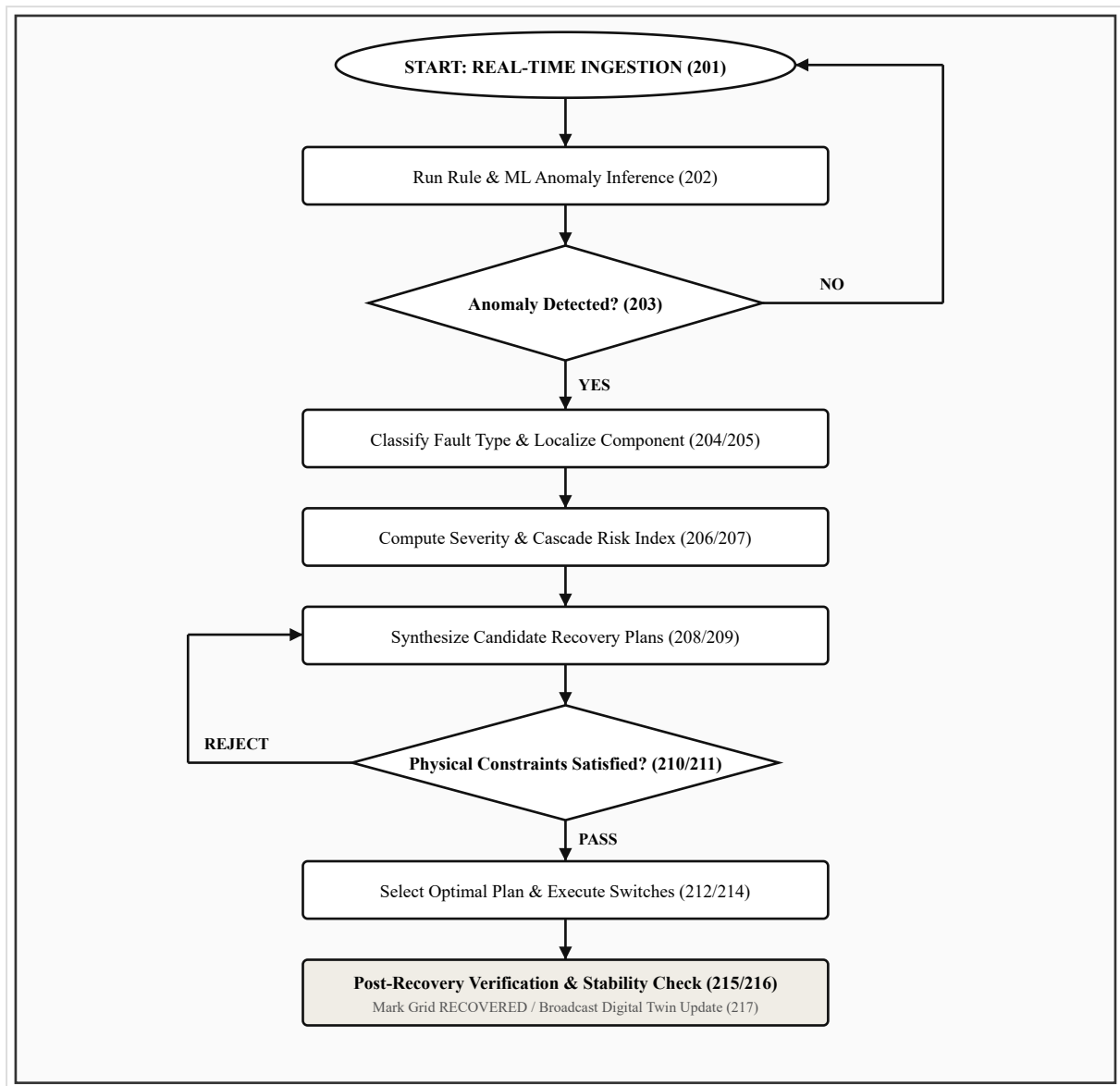


FIG. 2

Flowchart of Autonomous Self-Healing Pipeline

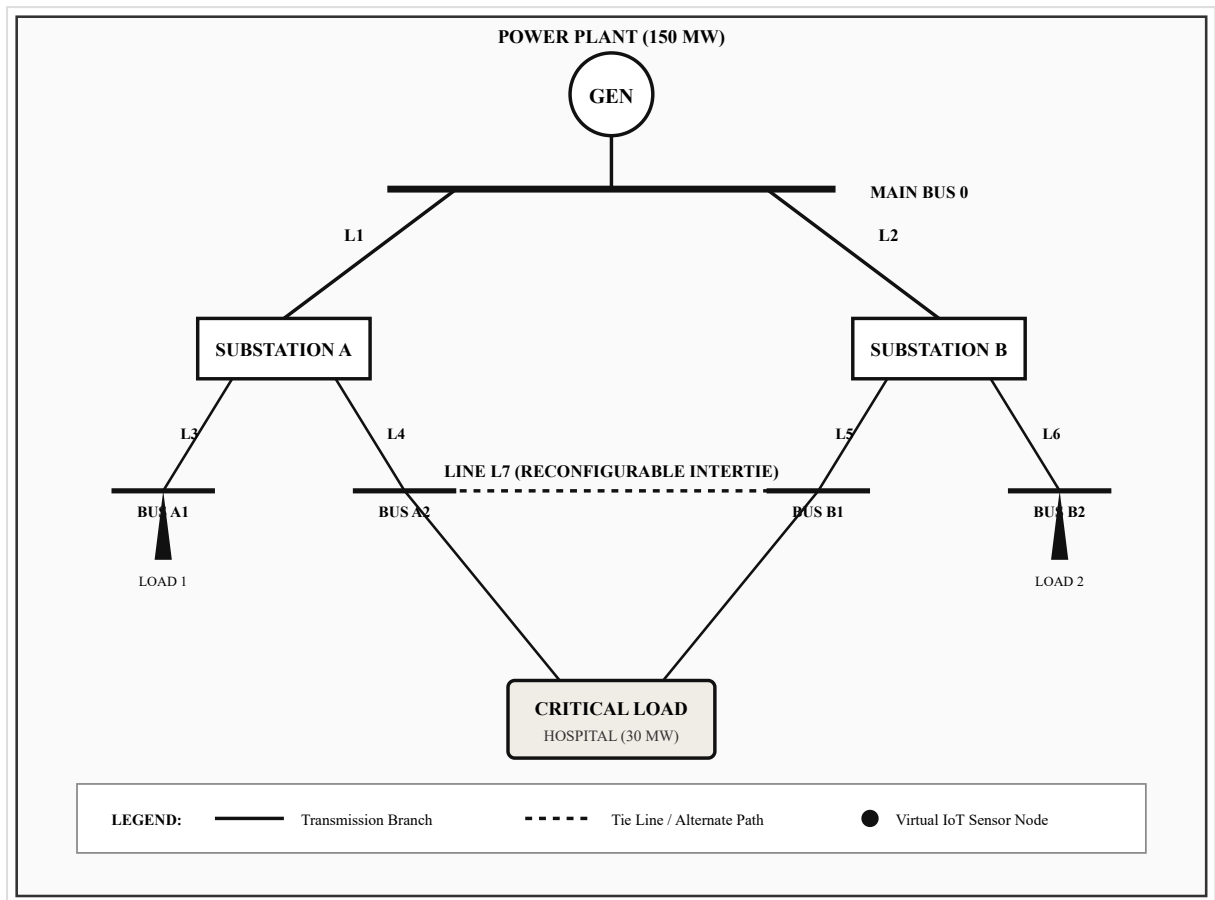


FIG. 3

Virtual Power Grid Single-Line Schematic with Redundant Alternate Routes

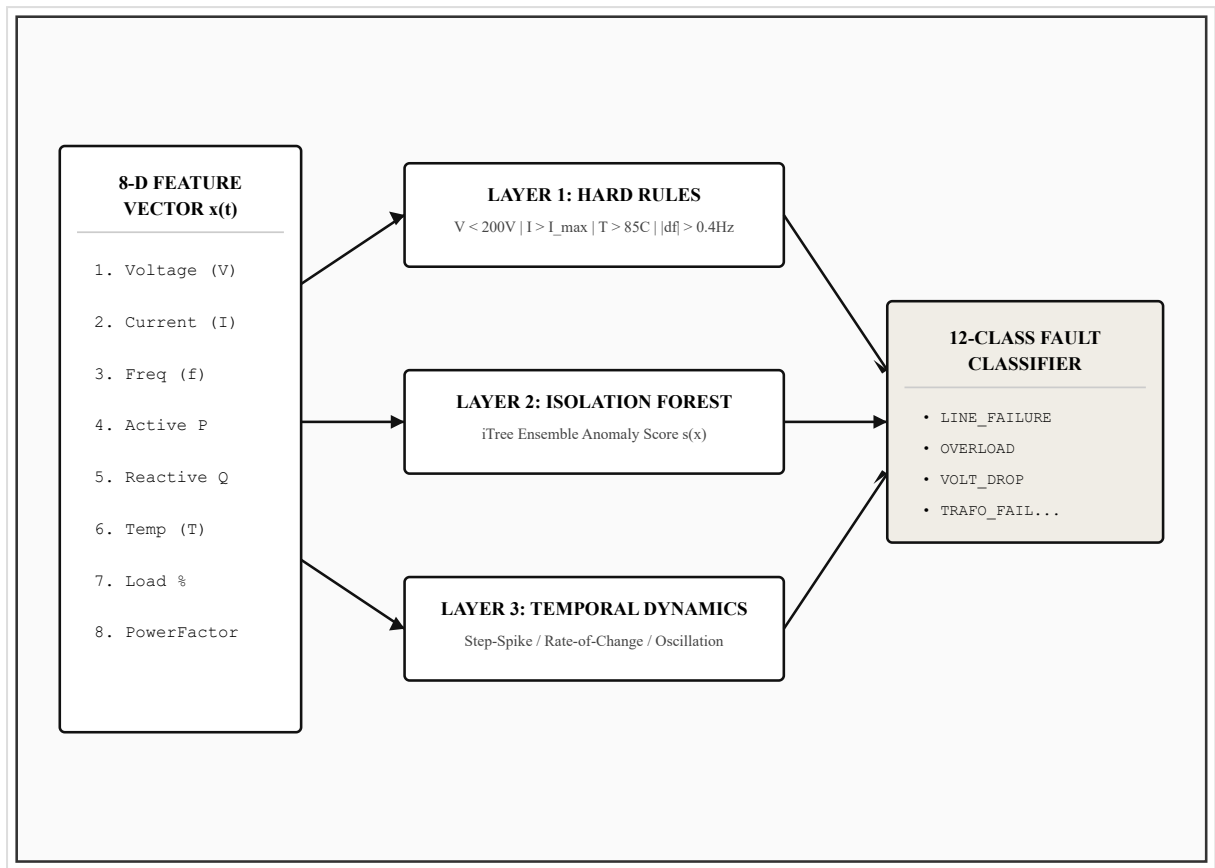


FIG. 4

Multi-Layer Anomaly Detection & Fault Classification Engine Architecture

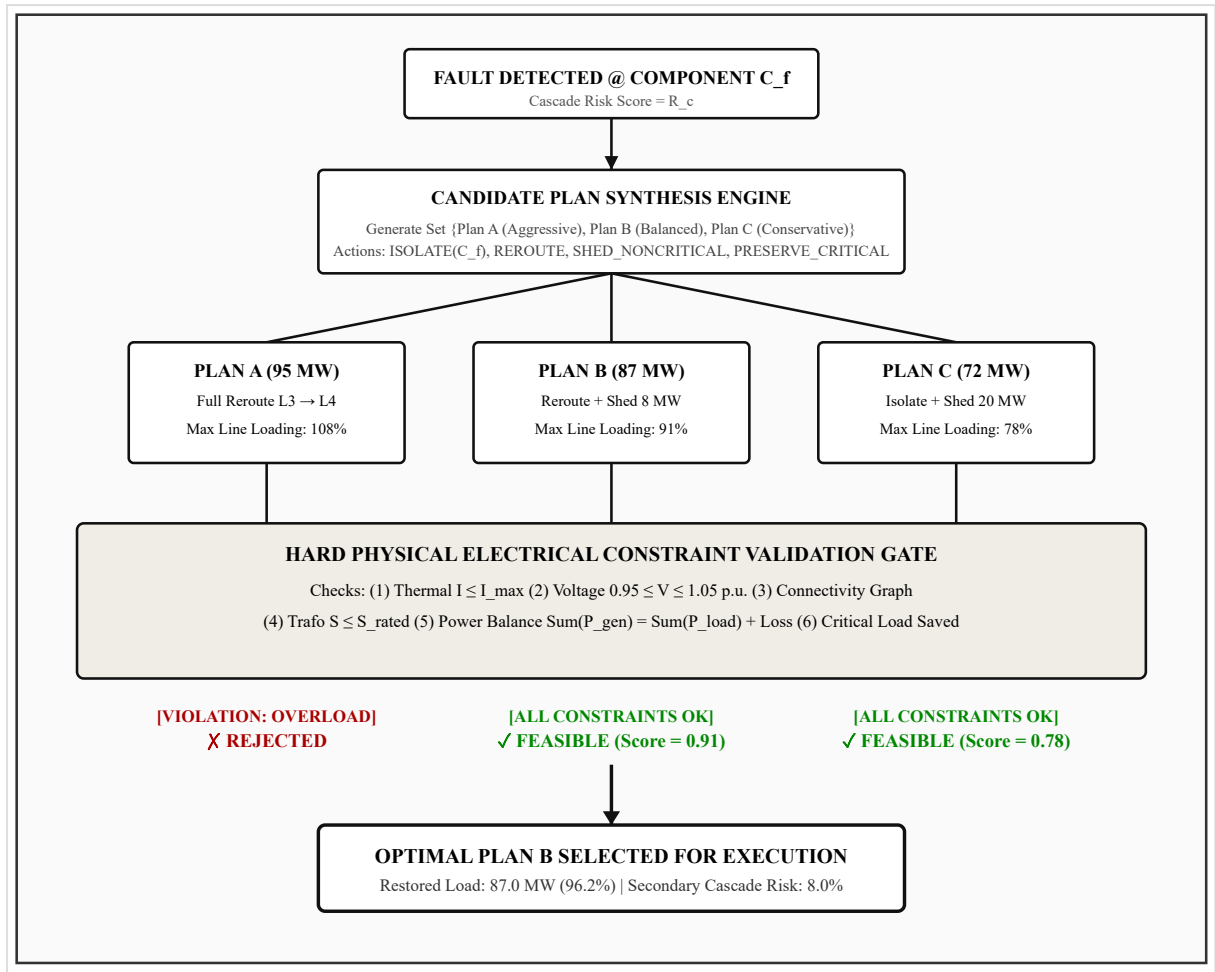


FIG. 5

Constraint-Aware Recovery Optimization Process and Multi-Objective Selection Gate

